```
!pip install transformers datasets torch scikit-learn pandas
```

```
equirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
equirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
equirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
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equirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1
equirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
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equirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=20
equirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
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equirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.
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equirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->dataset
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equirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
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equirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fs
equirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1
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equirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-
equirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->h
equirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.
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equirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<
```

```
import torch
from transformers import AutoTokenizer, AutoModelForSequenceClassification, Trainer, TrainingArguments
from datasets import load_dataset
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
import numpy as np
```

```
dataset = load_dataset("imdb")

print(dataset)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
```

README.md: 7.81k/? [00:00<00:00, 188kB/s]

plain_text/train-00000-of-00001.parquet: 100% 21.0M/21.0M [00:00<00:00, 105MB/s]

plain_text/test-00000-of-00001.parquet: 100% 20.5M/20.5M [00:00<00:00, 102MB/s]

plain_text/unsupervised-00000-of-00001.p(…): 100% 42.0M/42.0M [00:00<00:00, 109MB/s]

Generating train split: 100% 25000/25000 [00:00<00:00, 83628.70 examples/s]

Generating test split: 100% 25000/25000 [00:00<00:00, 108054.28 examples/s]

Generating unsupervised split: 100% 50000/50000 [00:00<00:00, 124394.53 examples/s]

```
DatasetDict({
    train: Dataset({
        features: ['text', 'label'],
        num_rows: 25000
    })
    test: Dataset({
        features: ['text', 'label'],
        num_rows: 25000
    })
    unsupervised: Dataset({
        features: ['text', 'label'],
        num_rows: 50000
    })
})
```

```
print(dataset['train'][0])
```

```
{'text': 'I rented I AM CURIOUS-YELLOW from my video store because of all the controversy that surrounded it when it was fir
```

```
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
```

config.json: 100% 483/483 [00:00<00:00, 45.5kB/s]

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 3.63kB/s]

vocab.txt: 232k/? [00:00<00:00, 6.94MB/s]

tokenizer.json: 466k/? [00:00<00:00, 14.4MB/s]

```
def tokenize_function(example):
    return tokenizer(example["text"], padding="max_length", truncation=True)
```

```
tokenized_datasets = dataset.map(tokenize_function, batched=True)
```

Map: 100% 25000/25000 [00:30<00:00, 824.77 examples/s]

Map: 100% 25000/25000 [00:37<00:00, 808.85 examples/s]

Map: 100% 50000/50000 [01:00<00:00, 889.97 examples/s]

```
tokenized_datasets = tokenized_datasets.remove_columns(["text"])
tokenized_datasets = tokenized_datasets.rename_column("label", "labels")
tokenized_datasets.set_format("torch")
```

```
print(tokenized_datasets["train"][0])
```

```
        0,     0,     0,     0,     0,     0,     0,     0,     0,     0,
        0,     0]), 'token_type_ids': tensor([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
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    0, 0, 0, 0, 0, 0, 0, 0]), 'attention_mask': tensor([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
    1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
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    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
small_train_dataset = tokenized_datasets["train"].shuffle(seed=42).select(range(2000))
small_test_dataset = tokenized_datasets["test"].shuffle(seed=42).select(range(1000))
```

```
model = AutoModelForSequenceClassification.from_pretrained("distilbert-base-uncased", num_labels=2)
```

```
model.safetensors: 100%                                    268M/268M [00:01<00:00, 264MB/s]

Loading weights: 100%                    100/100 [00:00<00:00, 372.77it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_transform.weight   | UNEXPECTED |
vocab_projector.bias     | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
vocab_layer_norm.weight  | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
classifier.bias          | MISSING    |
classifier.weight        | MISSING    |
pre_classifier.weight    | MISSING    |
pre_classifier.bias      | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream t
```

```
def compute_metrics(pred):
    labels = pred.label_ids
    preds = np.argmax(pred.predictions, axis=1)

    precision, recall, f1, _ = precision_recall_fscore_support(labels, preds, average='binary')
    acc = accuracy_score(labels, preds)

    return {
        'accuracy': acc,
        'f1': f1,
        'precision': precision,
        'recall': recall
    }
```

```
training_args = TrainingArguments(
    output_dir="./results",
    eval_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=2,
    weight_decay=0.01,
)
```

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=small_train_dataset,
    eval_dataset=small_test_dataset,
    compute_metrics=compute_metrics
)
```

```
text = "This movie was absolutely amazing!"

inputs = tokenizer(text, return_tensors="pt", truncation=True, padding=True)

outputs = model(**inputs)

pred = torch.argmax(outputs.logits)

if pred == 1:
    print("Positive Sentiment")
else:
    print("Negative Sentiment")
```

```
Positive Sentiment
```

```
import pandas as pd

# Evaluate the model
results = trainer.evaluate()

# Create table
data = {
    "Metric": ["Accuracy", "Precision", "Recall", "F1 Score"],
    "Score": [
        results["eval_accuracy"],
        results["eval_precision"],
        results["eval_recall"],
        results["eval_f1"]
    ]
}

result_table = pd.DataFrame(data)

print(result_table)
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as tru
  super().__init__(loader)
```

[62/125 07:13 < 07:27, 0.14 it/s]